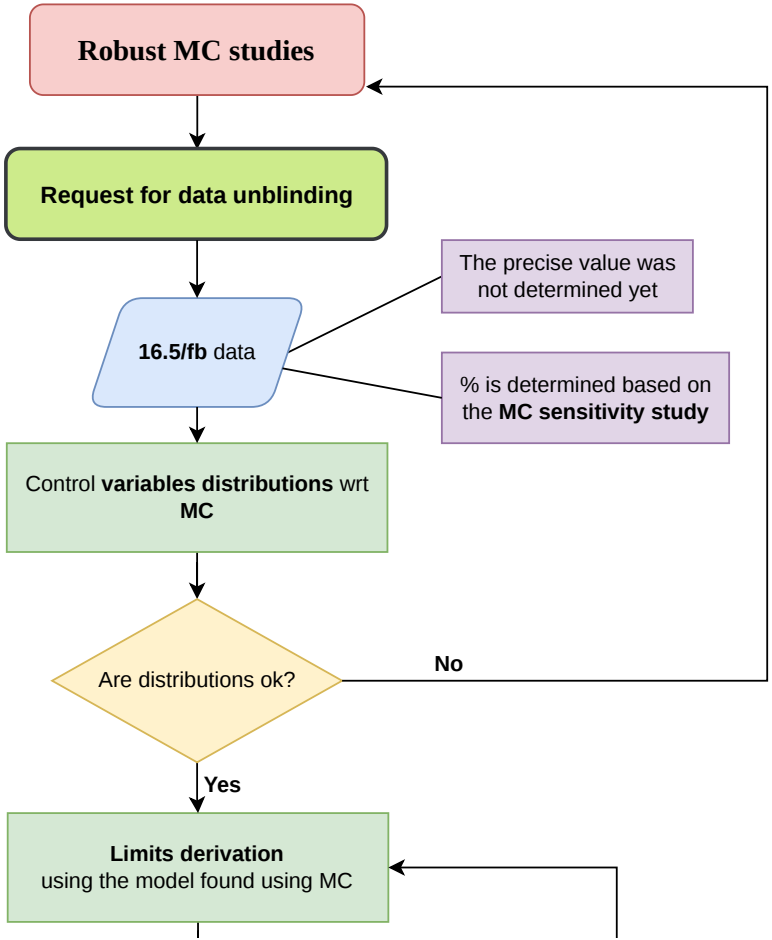
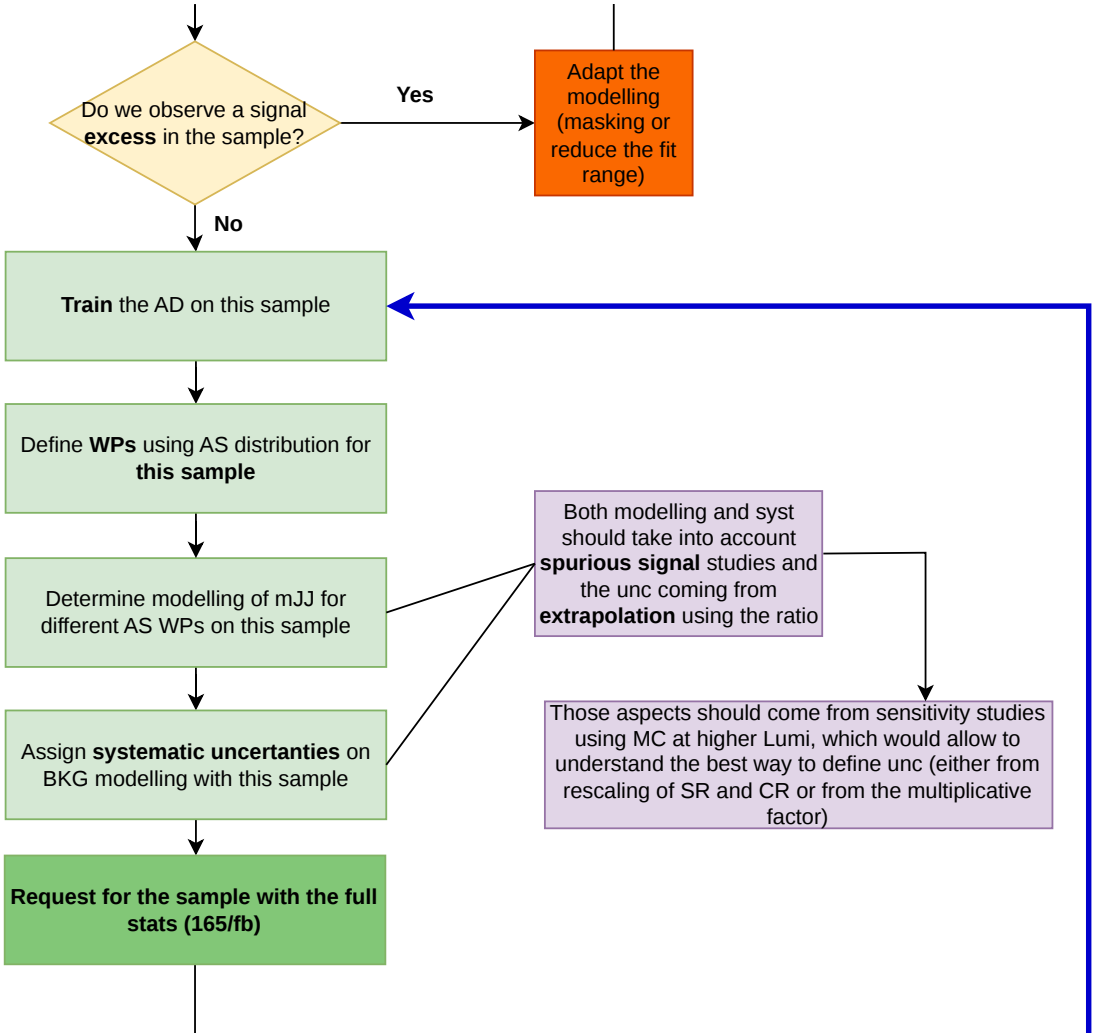
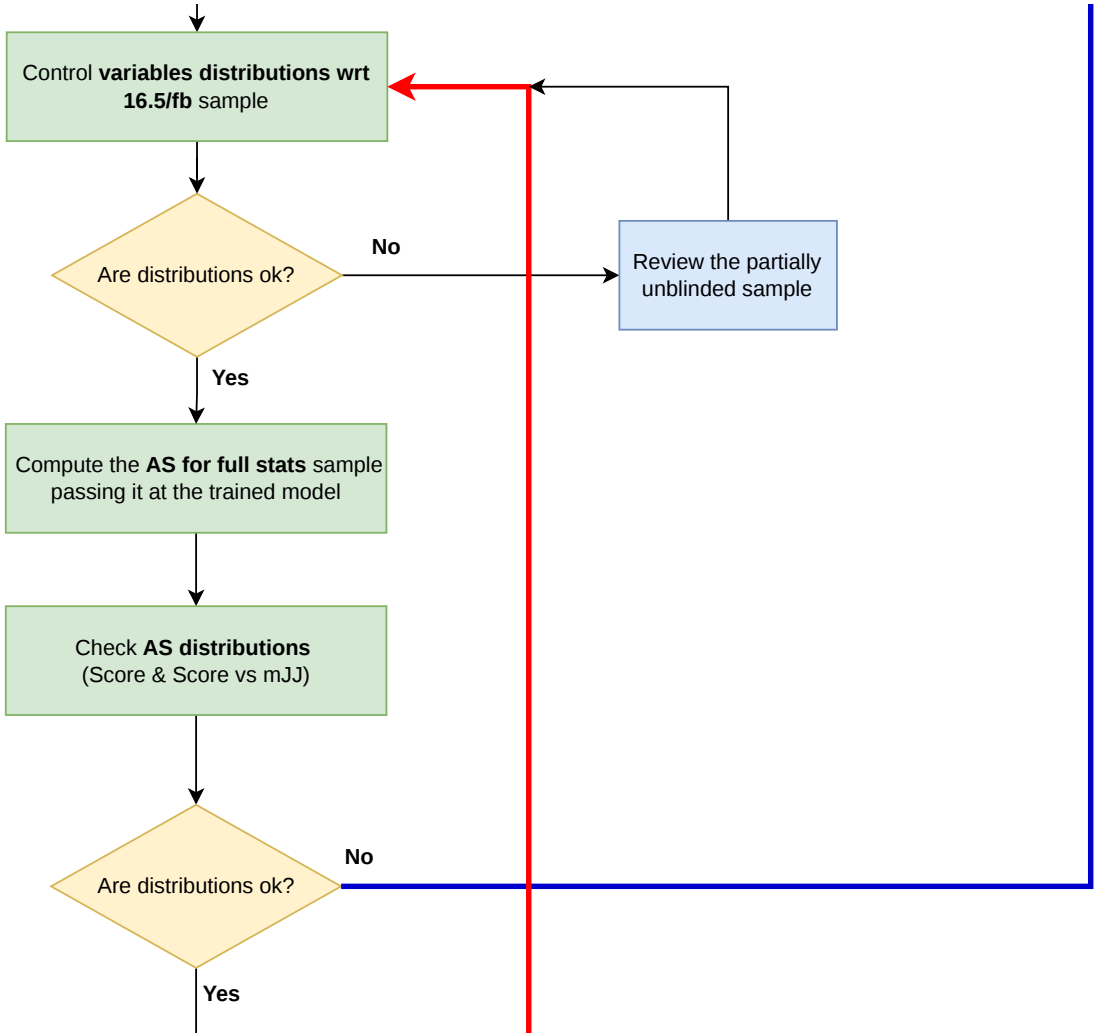
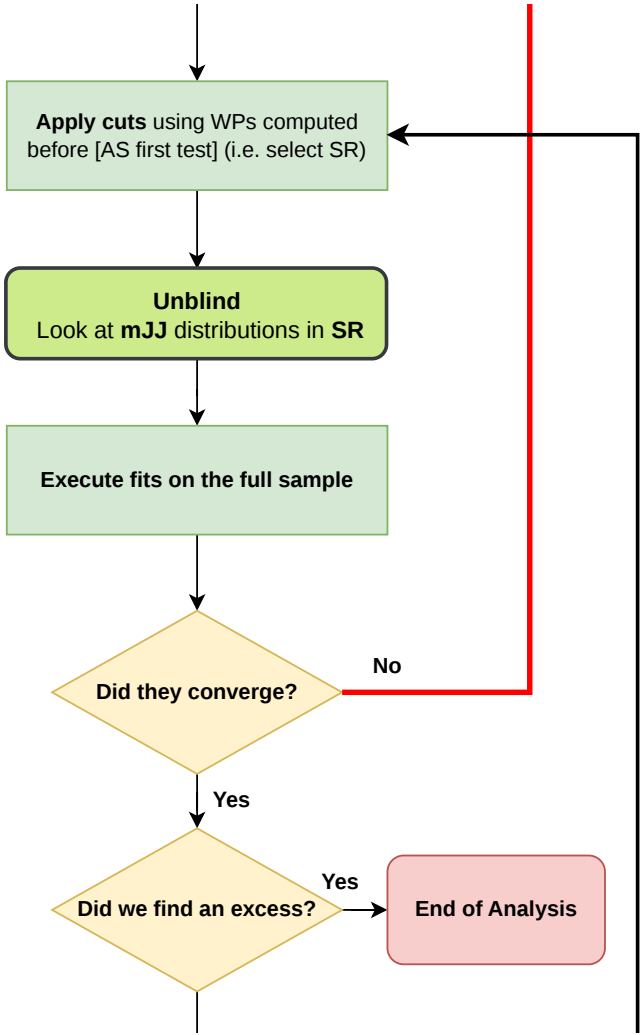
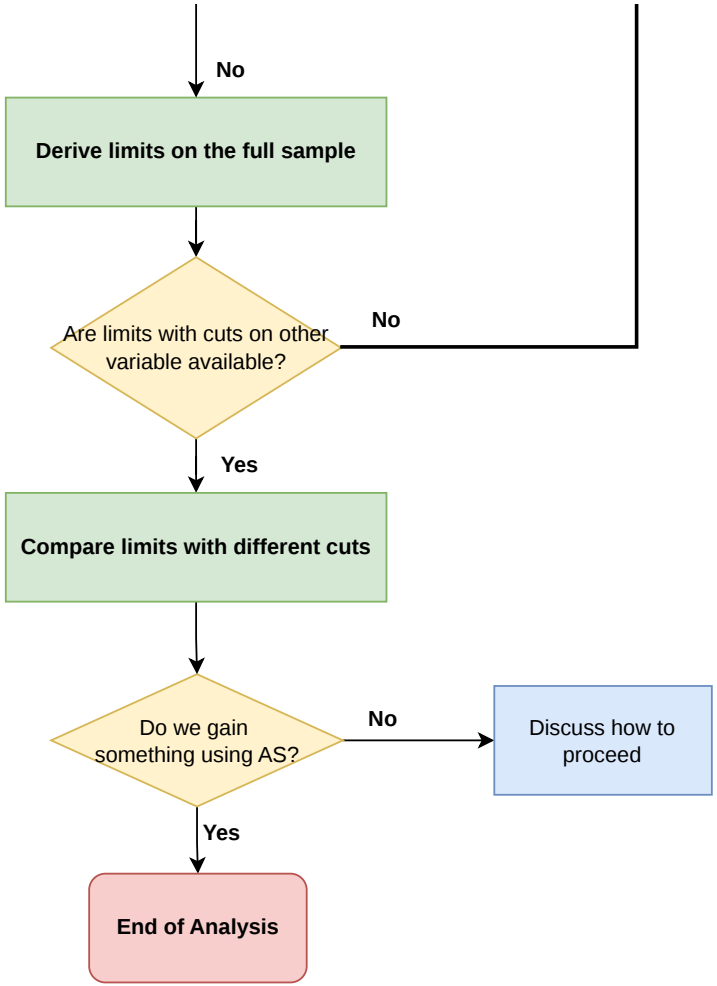
Analysis Strategy using data




Do we observe a signal **excess** in the sample?
Yes
Adapt the modelling (masking or reduce the fit range)
No
Train the AD on this sample
Define **WPs** using AS distribution for **this sample**
Determine modelling of mJJ for different AS WPs on this sample
Assign **systematic uncertanties** on BKG modelling with this sample
Request for the sample with the full stats (165/fb)
Both modelling and syst should take into account **spurious signal** studies and the unc coming from **extrapolation** using the ratio
Those aspects should come from sensitivity studies using MC at higher Lumi, which would allow to understand the best way to define unc (either from rescaling of SR and CR or from the multiplicative factor)


Control **variables distributions wrt 16.5/fb** sample
Are distributions ok?
No
Review the partially unblinded sample
Yes
Compute the **AS for full stats** sample passing it at the trained model
Check **AS distributions** (Score & Score vs mJJ)
Are distributions ok?
No
Yes


Apply cuts using WPs computed before [AS first test] (i.e. select SR)
Unblind
Look at mJJ distributions in SR
Execute fits on the full sample
Did they converge?
No
Yes
Did we find an excess?
Yes
End of Analysis


No
Derive limits on the full sample
Are limits with cuts on other variable available?
No
Yes
Compare limits with different cuts
Do we gain something using AS?
No
Discuss how to proceed
Yes
End of Analysis